DIEGO BARACHO RODOVALHO VARELA

Aerospace Engineering Student | Computational Fluid Dynamics and Aerodynamics
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Brazilian national, eligible for CERN graduate and student programmes (Brazil, Associate Member State since 13 March 2024)

PROFILE

Aerospace Engineering student specialising in computational fluid dynamics, with hands-on experience in finite-volume simulation, mesh generation and quality assessment, turbulence modelling and experimental validation in a wind tunnel. Led the aerodynamics and stability division of a university rocketry team through the design and flight of an award-winning hybrid-propulsion vehicle. Currently in a fifth semester of teaching assistantship, spanning aerodynamics, vector calculus and mechanics. My interest lies in fluid mechanics and numerical methods themselves rather than in a single application domain, and I am equally drawn to conjugate heat transfer, confined ventilation flows and non-continuum regimes as to external aerodynamics.

EDUCATION

B.Eng. in Aerospace Engineering

Universidade de Brasília (UnB), Gama Campus, Brasília, Brazil | June 2022 to December 2027 (expected, semester 2027.2)

- Final-year thesis (*Trabalho de Conclusão de Curso*) beginning in semester 2027.1, to be developed in OpenFOAM.
- Preparatory work assigned by my supervisor: introductory 3D external-flow simulations in OpenFOAM on the NASA Common Research Model wing-body geometry (AIAA Drag Prediction Workshop, DPW7), a first exercise in 3D case setup and meshing.
- Core coursework in fluid mechanics, aerodynamics of aerospace systems, heat transfer, structural mechanics, mechanical vibrations, vector calculus, differential equations and numerical methods.

TECHNICAL SKILLS

CFD solvers	Ansys Fluent (2D airfoil RANS, external flow over launch vehicles, URANS, DDES); OpenFOAM (case structure, blockMesh, snappyHexMesh, fvSchemes and fvSolution configuration, pressure/velocity coupling algorithms, turbulence models, post-processing)
Simulation setup	Turbulence model selection and understanding of the underlying formulation; inviscid and viscous modelling; boundary condition definition; mesh generation and refinement, with interpretation of y+ and mesh quality parameters; adapting solver configuration to the physics of each problem
Flight dynamics	OpenRocket, RocketPy: trajectory simulation, apogee prediction, static margin and stability analysis
Programming	MATLAB; Python (numerical solver implementation); Linux command-line workflow
Manufacturing	Composite hand lay-up of glass and carbon fibre with epoxy resin; spray-gun painting and progressive hand sanding to controlled surface roughness; fabrication and assembly of test hardware
Experimental	Wind tunnel testing; aerodynamic balance development and measurement validation; surface roughness characterisation
CAD and modelling	CATIA; aerodynamic surface design and modelling of fins, nose cones and boat tails
Domains	Aerodynamics; fluid mechanics; heat transfer; turbulence modelling; drag reduction; rocket stability

ENGINEERING AND RESEARCH EXPERIENCE

Head of the Aerodynamics and Stability Division

Capital Rocket Team, Universidade de Brasília | June 2024 to November 2025 | Faculty advisor: Prof. Olexiy Shynkarenko

- Technical lead of the division responsible for the aerodynamic design and stability analysis of the team's launch vehicle.
- Coordinated the aerodynamic development of **Boscov**, a hybrid-propulsion rocket designed for a 3 km apogee.
- Trajectory and stability simulation in RocketPy, refining apogee and static margin predictions for the vehicle.
- Hands-on fabrication of the vehicle's aerodynamic structures by composite hand lay-up of glass and carbon fibre with epoxy resin.
- Researched paint application methods for skin friction drag reduction and carried out the resulting process, applying the coating with a pressurised spray gun and progressively sanding every aerodynamic surface by hand to reach a measured roughness below 5 μm.
- Trained and technically supervised new members; standardised the division's simulation methodologies and workflows.
- Boscov was launched at the **Latin American Space Challenge (LASC) 2025** in Iacanga, São Paulo, and took **first place in the Hybrid/Liquid Rockets category**.

Safety Manager, Boscov Mission

Capital Rocket Team, Latin American Space Challenge (LASC) | 2025

- One of the four mandatory positions in the mission structure required by the competition rules: the mission's primary authority on safety, accountable for compliance with the LASC Safety Policy across design, development, testing and on-site operations.
- Responsible for the mission's Failure Modes and Effects Analysis (FMEA) and risk assessments, its safety reviews, and the response to nonconformities during the event.

Member, Aerodynamics and Stability Division

Capital Rocket Team, Universidade de Brasília | December 2023 to June 2024

- Trajectory simulation and stability analysis in OpenRocket, evaluating apogee, static margin and in-flight vehicle behaviour.
- External flow CFD simulations around the launch vehicle in Ansys Fluent, focused on aerodynamic coefficient estimation and drag characterisation.
- Design and modelling of the vehicle's aerodynamic components: fins, nose cone and boat tail.

Aerodynamic Balance Development: Collaboration Between Teams

Capital Rocket Team with Draco Volans and Mamutes do Cerrado (UnB aerodesign teams) | First half of 2024

- Contributed to the structural design of an aerodynamic balance for force and moment measurement, and to the fabrication of part of its components.
- Validated balance readings in the wind tunnel of UnB's Faculty of Technology (FT), testing flat plates across a range of angles of attack and correlating the measured loads with analytical predictions.

INDEPENDENT PROJECT

Turbulence Modelling Study: URANS versus DDES

Self-directed | *Ansys Fluent*

- Compared URANS (SST k-ω) with delayed detached-eddy simulation for the flow over a NACA 4412 aerofoil at 14° incidence, just before stall: Re $\approx 9 \times 10^6$, 780k cells, maximum y+ ≈ 1.72.
- URANS lift and drag coefficients agreed closely with published experimental data, while DDES overpredicted them by 30% to 40%, a discrepancy I attributed to mesh resolution insufficient for a scale-resolving method within the one-million-cell limit of the Ansys Student licence. Written up as a technical post on LinkedIn.

TEACHING EXPERIENCE

Teaching Assistant, Aerodynamics of Aerospace Systems

Universidade de Brasília | *September 2025 to present* | *Supervisor: Prof. Olexiy Shynkarenko*

- Guidance on 2D airfoil CFD simulations with RANS turbulence modelling in Ansys Fluent, covering pre-processing, solution and post-processing.
- Structured mesh generation and refinement, with assessment of quality metrics (y+, aspect ratio, skewness) and boundary condition definition.
- Turbulence model selection, numerical scheme configuration and convergence assessment through residual analysis.
- Extraction and interpretation of lift and drag coefficients, and of pressure and velocity contours over the airfoil.
- Flow field analysis to identify boundary layer separation and the onset of stall, guiding students in reading the physics behind the numerical results.
- Course taught on the basis of John D. Anderson's *Fundamentals of Aerodynamics*.

Teaching Assistant, Calculus III (Multivariable and Vector Calculus)

Universidade de Brasília | *2024*

- Student support in double and triple integrals, coordinate transformations, line and surface integrals, vector fields, and the Green, Stokes and divergence theorems: the integral theorems underlying the conversion between integral and differential forms of the conservation equations.

Teaching Assistant, Physics I (Classical Mechanics)

Universidade de Brasília | *2023*

- Student support on problem sets, based on Moysés Nussenzveig's *Curso de Física Básica*: kinematics, Newtonian dynamics, work and energy, momentum, rigid body dynamics and oscillations.

AWARDS AND DISTINCTIONS

- **First Place, Hybrid/Liquid Rockets Category.** Latin American Space Challenge (LASC) 2025, Iacanga, Brazil. Awarded to the Boscov project, whose aerodynamic development I led.

PROFESSIONAL TRAINING

- **Computational Fluid Dynamics Fundamentals, Part 1** (Aidan Wimshurst, Fluid Mechanics 101; Udemy): finite-volume discretisation of the scalar transport equation; central versus upwind differencing; Péclet number and stability; solver implementation from first principles.
- **Computational Fluid Dynamics Fundamentals, Part 2** (Aidan Wimshurst, Fluid Mechanics 101; Udemy): 2D formulations; Dirichlet and Neumann boundary conditions in matrix form; residuals and heat flux balances as verification criteria; wall functions (y+, u+, y*), including temperature wall functions.
- **Computational Fluid Dynamics Fundamentals, Part 3** (Aidan Wimshurst, Fluid Mechanics 101; Udemy): unstructured mesh discretisation; cell volumes, face areas and unit normals for skewed and irregular cells; mesh quality metrics.
- **OpenFOAM Beginner Core Courses** (Caesar Wiratama; Udemy): case folder structure; blockMesh and snappyHexMesh; geometry preparation; boundary conditions; numerical schemes and pressure/velocity coupling; turbulence modelling; Linux workflow.
- **CATIA course** (Zenit Aerospace), 2022.

LANGUAGES

Portuguese	Native
English	Full professional proficiency. EF SET Certificate, 81/100 (C2 Proficient)
Spanish	Intermediate
Japanese	Elementary
Latin	Basic reading knowledge

RESEARCH INTERESTS

Finite-volume methods and discretisation schemes; turbulence modelling and near-wall treatment; mesh quality and its effect on solution reliability; conjugate heat transfer; confined and ventilation flows; rarefied and molecular flow regimes where continuum assumptions break down, as found in high-vacuum systems; correlation between CFD predictions and experimental measurement.

INTERESTS

Physics beyond the engineering syllabus, particularly mechanics, thermodynamics, electromagnetism and modern physics. Guitar for ten years, from popular music and rock to classical repertoire and flamenco; volunteered at a children's day-care school fundraiser in Paulista, Pernambuco, playing guitar for the children's singing. Competitive swimming through secondary school; Shotokan karate as a child, in kata and kumite.